STORAGE VS. THE FLEXIBLE LOAD: THE REFEREE'S FIRST QUESTION, ANSWERED

TEAM A1 — ROLE 1

Abstract. The first question a referee asks about a "computing load absorbs renewable surplus and that is good for carbon" story is: what about storage? Storage absorbs the same surplus and *returns* it to displace fossil, whereas the load consumes it for computing and displaces nothing on the grid. So the "no load" counterfactual over-credits the load. We concede the point in full and then show it does not damage the paper's results, because it is aimed at a claim the paper does not need. Three facts, all verified: (i) for pure surplus absorption storage lowers grid carbon and the load does not—storage dominates the load *as a carbon technology*; (ii) the load's carbon-relevant object is not surplus absorption but *dispatch*, and the naive-versus-surplus-contract carbon gap is exactly invariant to how much storage is installed; (iii) the load is the higher-*value* user of a surplus MWh precisely when computing value clears a sharp threshold $R > c + \eta\,[\varphi^{\circ}_{\mathrm{dm}} c + (1-\varphi^{\circ}_{\mathrm{dm}})\beta + p]$, robustly met by high-value compute—and, crucially, a *higher* tariff (the empirical correction the calibration critics demand) only lowers storage's value and pushes the verdict *toward* the load. The load is not a carbon mitigation technology; it is a computing demand whose dispatch is a carbon lever, and that lever is orthogonal to storage.

1. The model with both assets

Keep the residual-demand primitive $x = D - K_R\tilde{S}$ with c.d.f. $F(\cdot; K_R)$. In a surplus state ($x < 0$) the curtailable surplus is $(-x)$. Two assets can take it. Storage of capacity K_S and round-trip efficiency $\eta \in (0,1)$ charges $\min((-x), K_S)$ and returns η of it in later peak states, *displacing fossil*. The flexible load of capacity K_M computes on surplus, earning the margin $R - c$ per MWh and *displacing no grid fossil*. Storage is the fossil-displacer, so we let it charge first; the load takes the residual surplus $((-x) - \text{charge})_+$. Grid carbon (fossil burned) is

$$\mathbb{E}[\text{fossil}] \;=\; \underbrace{\mathbb{E}[\min(x_+, K_F)]}_{\text{consumer fossil}} \;+\; \underbrace{\mathbb{E}[\text{naive load burn}]}_{=0 \text{ under surplus contract}} \;-\; \underbrace{\eta\,\mathbb{E}[\min((-x)_+, K_S)]}_{\text{storage displacement}}. \tag{1}$$

2. Storage dominates the load as a carbon technology

Proposition 1 (Per-MWh carbon: the referee is right)**.** *A unit of surplus routed to storage lowers grid fossil by η times the fraction it returns into fossil-serving states; the same unit routed to the (surplus-contracted) load lowers grid fossil by* zero*. Storage strictly dominates the load in grid-carbon terms for surplus use.*

Proof. Immediate from (1): K_S enters with a strictly negative sign, K_M (surplus contract, naive burn $= 0$) does not enter at all. The load computes; it does not return energy to the grid. □

Numerically, at $K_R = 2, K_F = 7, \eta = 0.85$: installing $K_S = 3$ cuts grid fossil from 3.9333 to 3.9207 (-0.0126); installing a surplus-contracted $K_M = 3$ leaves it at 3.9333 (-0.0000). *The consequence is that the "no load" counterfactual of Week 3 overstates the load's carbon benefit*: the honest benchmark credits storage with the fossil it would have displaced, and against that benchmark the surplus load is carbon-neutral, not carbon-negative.

Date: Addressing the missing substitute.

3. The dispatch gap is storage-invariant

The load's carbon-relevant object is not whether it exists but how it runs. Under the *naive* rule it burns fossil for computing (Week 2's floor); under the *surplus contract* it does not. That difference is what Week 3 and Week 4 are about, and it is untouched by storage.

Proposition 2 (The naive−surplus carbon gap does not depend on K_S)**.** *Let* $\Delta_{\text{disp}}(K_S) = \mathbb{E}[\textit{fossil} \mid \textit{naive}, K_S] - \mathbb{E}[\textit{fossil} \mid \textit{surplus}, K_S]$. *Then* $\Delta_{\text{disp}}(K_S) = \mathbb{E}[\textit{naive load burn}]$ *for all* K_S*: the storage term in* (1) *is common to both dispatch modes and cancels.*

Proof. Both modes share the same consumer fossil and the same storage displacement $-\eta\mathbb{E}[\min((-x)_+, K_S)]$; they differ only in the naive burn term, which is present under naive dispatch and zero under the surplus contract. Subtracting, the storage term cancels identically. □

Verified: at $K_S = 0, 2, 4$ the gap is $+2.1291$ in every case, constant to the reported digits.

Corollary 3 (The gap survives storage rivalry; the crowding is second order)**.** *Proposition 2 lets storage charge first, so the surplus load never cannibalizes storage. Grant instead the load* co-priority*—it takes surplus up to* K_M *and storage gets only the residual. Then the surplus contract does crowd out some storage displacement, and the gap becomes* $\mathbb{E}[\textit{naive burn}] - \eta\big(\mathbb{E}[\min((-x)_+, K_S)] - \mathbb{E}[\min(((-x)_+ - K_M)_+, K_S)]\big)$*. The crowding term is* second order*: the avoided naive burn accrues in* fossil-serving *states (where the naive load runs on fossil), while the crowded-out storage charge is lost in* surplus *states—disjoint events. Numerically the gap falls only from* $+2.129$ *to* $+2.116$ *(a* 0.6% *reduction) as* K_S *ranges up to* 10*: the crowded-out displacement is* $\eta \times 0.013$*, two orders of magnitude below the avoided burn* 2.129*. So the dispatch result is robust to the rivalry, not an artifact of the charging order.*

So every carbon result the paper actually proves—the naive rule raises carbon (Week 4), the surplus contract is its carbon-optimal dispatch (Week 3)—holds with storage in the model, and holds whether or not the load competes with storage for the surplus. Storage changes the *level* of carbon, not the *dispatch comparison.*

4. Who should take the surplus: a value threshold

Storage wins the carbon comparison; it does not follow that storage should take all the surplus, because the two assets sell different products. Storage arbitrages energy; its expected value per charged MWh is η times the marginal displacement price—generation cost c in the common case, the outage price $\beta + p$ in the rare shortfall—so $v_{\text{stor}} = \eta\,[\varphi^\circ_{\text{dm}} c + (1 - \varphi^\circ_{\text{dm}})\beta + p]$ using the *main-model* fossil cutoff $F(K_F) = \varphi^\circ_{\text{dm}} = (\beta + p - c - C_F)/\beta$ of the double-margin note. (Using this cutoff, not the utility-operated φ°, is what makes the value comparison consistent with the paper's actual model—an inconsistency worth flagging, since it moves the verdict.) The load produces computing at margin $R - c$.

Proposition 4 (The load is the higher-value surplus user iff computing clears a threshold)**.** *The flexible load is the higher-value user of a marginal surplus MWh iff*

$$\boxed{R \;>\; c + \eta[\varphi^\circ_{\text{dm}} c + (1 - \varphi^\circ_{\text{dm}})\beta + p] \;=:\; R^\star.}$$

Below $R^\star$ *storage should take the surplus; above it the load should. Both are used at an interior joint optimum whenever the surplus exceeds either asset's capacity, so they are* complements sorted by the load's value*, not pure substitutes.*

At the calibration ($c = 0.5, \eta = 0.85, \beta + p = 4.7, C_F = 0.6$, so $\varphi^\circ_{\text{dm}} = 0.90$) the threshold is $R^\star = 1.28$, and $v_{\text{stor}} = 0.782$. The paper's own $R = 1.4$ (hence $R - c = 0.90 > 0.782$) *clears* it:

at the calibration the flexible load *is* the higher-value surplus user, and storage should take the surplus only for a lower-value load. The verdict is margin-sensitive—plugging the utility-operated $\varphi^\circ = 0.857$ instead would give $R^\star = 1.44$ and reverse it—which is exactly why the value comparison must be run in the main model. It also depends on how rare outage is:

$\Pr(\text{outage}) = 1-\varphi^\circ$	v_{stor}	vs $R-c=0.90$	higher value
0.02	0.50	$>$	load
0.05	0.60	$>$	load
0.10	**0.78**	$>$	**load ← main model ($\varphi^\circ_{\text{dm}} = 0.90$)**
0.14	0.93	$<$	storage ← utility-op ($\varphi^\circ = 0.857$)
0.20	1.14	$<$	storage

The main-model row ($\Pr(\text{outage}) = 1-\varphi^\circ_{\text{dm}} = 0.10$) is the relevant one, and there the load clears; the utility-operated row (0.14) is the mismatched comparison that reverses it.

The verdict is robust, not knife-edge: both contested inputs push toward the load. A fair worry is that "the load clears" rests on a thin margin and a disputed tariff (EIA-861 rebuilds p at \$66–90 versus the assumed \$50). Two facts dissolve the worry, and both cut the *same* way.

Higher p helps the load. Storage's value $v_{\text{stor}} = \eta[\varphi^\circ_{\text{dm}} c + (1-\varphi^\circ_{\text{dm}})\beta + p]$ *falls* as p rises, because a higher tariff pushes $\varphi^\circ_{\text{dm}} = (\beta + p - c - C_F)/\beta$ toward one, making outages rarer and storage's outage-arbitrage premium scarcer: $v_{\text{stor}} = 0.935, 0.782, 0.612$ at $p = 0.5, 0.7, 0.9$. (The interior cutoff requires $\varphi^\circ_{\text{dm}} < 1$, i.e. $p < c + C_F = 1.1$; at $p = 1.1$ the equilibrium is a degenerate zero-outage corner and beyond it the interior formula no longer applies, so we stop the band at $p = 0.9$.) So the empirical correction the calibration critics demand—a *higher* tariff—moves the verdict *toward* the load, not away from it.

Only the doubly-conservative corner favors storage. The load's surplus value $R - c$ is independent of p, so the verdict is a clean function of (R, p) over the valid band:

$R\backslash p$	0.5	0.7	0.9
1.4	storage	load	load
1.8	load	load	load
2.5	load	load	load
4.0	load	load	load

The single cell where storage wins is $R = 1.4$, $p = 0.5$—the lowest computing value *and* the lowest tariff at once. Bitcoin and AI compute sit far above $R = 1.4$ (that value is a deliberately conservative floor); for any $R \geq 1.8$ the load is the higher-value surplus user at *every* tariff in the band, clearing by a margin of $+1.06$ at $R = 2.5$ and $+6.56$ at $R = 8$ against the worst-case $v_{\text{stor}} = 0.935$ over the valid band $p \in [0.5, 0.9]$. The distance to the boundary grows with computing value; the verdict is a coin-flip only at the corner where the load is barely worth running, and there the answer rightly tips to storage. Pinning p against EIA-861 (an A2 task) sharpens the number but cannot flip a verdict that a higher p only reinforces.

5. What this settles

Remark 5 (The honest bottom line). Storage is the better carbon technology; the flexible load is not a carbon technology at all. Its carbon footprint in the model is a *dispatch* property, and the paper's results are dispatch results: the naive rule raises emissions, the surplus contract is the carbon-optimal way to run a load that exists for computing, and that comparison is exactly invariant to storage (Proposition 2). What the storage counterfactual correctly kills is the weaker reading that a surplus-absorbing load is carbon-*negative* versus doing nothing—against the right

benchmark (storage) it is carbon-neutral, and the paper should, and now does, benchmark the surplus contract against the naive rule rather than against a vacuum. The load earns its place on private value (Proposition 4), which high-value compute clears; the grid should still prefer storage for the surplus when the load's computing is cheap. This is a cleaner and more defensible claim than "computing greens the grid," and it is the one the model supports.

Limits. This is a static, expected-flow treatment of storage: it prices the returned energy at the expected marginal displacement, not through an explicit intertemporal state-of-charge process. A dynamic storage model with binding state-of-charge constraints would change the *magnitudes* v_{stor} and the displacement fraction, and could introduce a scarcity value when storage is small; it would not change Proposition 2 (the dispatch gap cancels the storage term whatever its size) nor the *existence* of a value threshold in Proposition 4. The full dynamic (renewable, fossil, storage, flexible-load) program is the natural next model and is out of Role 1's scope.